

Sensory Scaffolding to Reconnect Users with the Ordinary

Yujie Tao
Stanford University
Stanford, California, USA
yjtao@stanford.edu

Abstract

Digital technologies, such as social media and virtual reality, have transformed how people interact by enabling a sense of virtual presence. However, this often comes at the cost of drawing users away from real-world experiences. My research explores how technology can help users reconnect with ordinary, real-world interactions, fostering a stronger sense of "being here." Grounded in the understanding of the perceptual processes that lead to habituation and disengagement from the ordinary, I design sensory scaffolding—systems that temporarily modulate users' perceptual experiences to offer new perspectives on mundane interactions.

This manuscript provides an overview of my research on sensory scaffolding to support both physical and social interactions. I will introduce wearable systems that augment sounds produced by the user's hands to support mindfulness in manual interactions. Similarly, routine social encounters—such as passing by another person on the street—can be renewed through body-anchored audio cues. My ongoing work taps into everyday creativity, developing systems that visually augment familiar objects to encourage creative use. I will conclude with future work, including studying the long-term effects of sensory scaffolding systems and addressing safety and ethical considerations.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Multisensory Interfaces, Digital Well-being.

ACM Reference Format:

Yujie Tao. 2025. Sensory Scaffolding to Reconnect Users with the Ordinary. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3746058.3758461>

1 Introduction

Digital technologies—from social media to virtual reality—reshape human interaction by enabling experiences that transcend physical constraints. With just a few clicks, remote users can connect in

real time through online platforms or become immersed in three-dimensional virtual spaces—evoking a sense of virtual presence, i.e., "being there." Despite the benefits, an overemphasis on presence in the digital world can come with downsides. For instance, instant feeds on social media often pull users' attention away from their immediate physical and social surroundings, contributing to mental distress and social isolation [5, 18]. Head-mounted displays intensify this tendency by directly obstructing users' vision with external hardware, creating immersion by detaching users from the present moment [1].

Beyond striving for virtual experiences that evoke a sense of "being there," **my research explores how to design interfaces to help users reconnect with real-world, ordinary interactions—fostering a sense of "being here."** This goal echoes a long-standing aspiration in human-computer interaction, tracing back to foundational work on ubiquitous computing [27, 28]. By "ordinary," I mean the routine and often overlooked moments of daily life, as simple as commuting to work, passing someone on the street, or cooking a meal. While these moments may seem small, they can offer relief from everyday stress when approached mindfully [7], foster meaningful social connections [6], and even spark the "aha" insights that contribute to creative discovery [20].

Why is engagement with the ordinary difficult to achieve? It stems from the repetitive, predictable nature of everyday interactions. From a Bayesian perspective, the human brain continuously predicts sensory input based on prior experiences and updates these predictions when incoming information deviates from expectations. In routine contexts, users are surrounded by familiar objects and people, and repeated exposure to the same stimuli can lead to habituation—a process in which sensory responses gradually diminish over time [25, 26]. As a result, users may become less attentive to everyday actions, cognitively fixed on the typical functions of familiar objects, or socially disengaged during routine encounters.

Existing systems designed to support "being here" often deliver cognitive strategies intended to shift how users think about their engagement with technology and the routine. For instance, these systems leverage reminders [15, 21], goal-setting [8], and reflective prompts or artifacts [12, 17]. While valuable, cognitive strategies can be abstract, making them difficult for users to internalize or act on, especially when routine demands already fragment users' cognitive resources. New solutions are needed to support users in the ordinary moments, beyond instructing them to "think differently".

I propose interfacing directly with the perceptual processes that shape how users interpret the ordinary, enabling users to "perceive differently." My Ph.D. research focuses on **sensory scaffolding—systems that temporarily modulate perceptual experiences to disrupt habituation and encourage a fresh perspective with everyday moments**. In the following sections, I will demonstrate wearable devices that provide subtle modulation of

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758461>

sounds produced by users' hands and how such devices can promote mindful interaction with everyday objects, rekindling attention and curiosity in manual tasks. Similarly, mundane social encounters can gain new meaning when augmented by body-anchored sounds triggered by simple movements, such as passing by another person. In my ongoing work, I explore how visual augmentation can inspire creative uses of everyday objects. Taken together, these investigations demonstrate new ways to support a sense of presence during everyday moments when cognitive resources are fragmented. Lastly, I will lay out future directions for sensory scaffolding systems, including studying long-term impact and designing adaptive mechanisms. I will also discuss design considerations to mitigate potential risks associated with modulating perceptual experiences.

2 Sensory Scaffolding to Support Mindfulness in Manual Interactions

Manual interactions are fundamental to everyday tasks. However, many of these routine actions—such as folding clothes or washing dishes—are often perceived as dull and uninteresting. As a result, people frequently perform them without being mindful about their actions. For example, one might be cooking while thinking about a recent conversation with a colleague. The tendency to unintentionally drift away from the present moment was found to reduce self-reported happiness [13, 22] and impair cognitive performance [16, 19]. This is where sensory scaffolding could offer support by temporarily modulating perceptual experiences to promote mindfulness: the state of maintaining attention to the present moment with curiosity and openness.

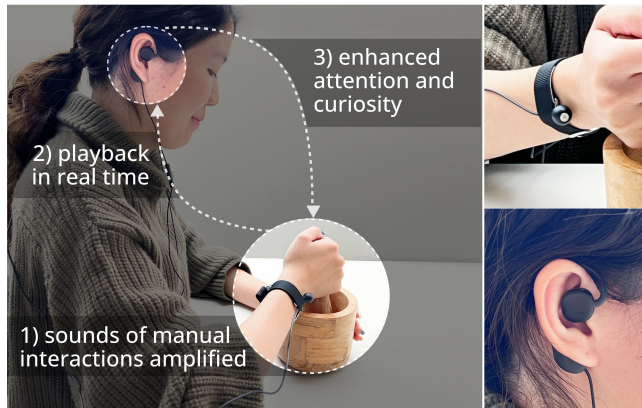


Figure 1: We propose a wearable device that amplifies sounds around the users' hands and plays them back to the user in real time, encouraging attention and curiosity toward the present-moment experience.

In particular, we introduced a wearable device that amplifies the sounds around the user's hands and plays them back to the user in real time. By enhancing the sounds produced by ongoing manual actions, the device increases the salience of these signals, which lowers the barrier for users to self-regulate their attention to these cues. Additionally, the subtle yet unexpected auditory variations introduced by the device create a form of Bayesian surprise—signals that slightly deviate from the sensory input users typically expect

from their own manual interactions. As a result, the device taps on two core components of mindfulness: attention and curiosity, as operationalized by Bishop et al [4].

While other features of sound—such as pitch, frequency, or reverberation—can also influence salience and introduce sensory surprise [9, 11], we chose to only modulate loudness because it preserves most of the perceptual qualities of existing sounds. This decision reflects our goal for the device to act as a temporary scaffold for mindfulness—not to replace it. When users stop using the device, they are still able to notice and relate to the same sounds that were previously amplified. Through this formative study, we co-design the preferred level of audio augmentation with two mindfulness coaches.

To examine the effect of the proposed device on mindfulness in everyday tasks, we conducted an in-lab study ($n = 60$). Participants were asked to explore a set of given objects. Using a mixed-factorial design, half of the participants experienced audio augmentation (*device group*), while the other half did not (*control group*). The study included two sessions: in one session, participants interacted with familiar objects; in the other, they explored unfamiliar objects.

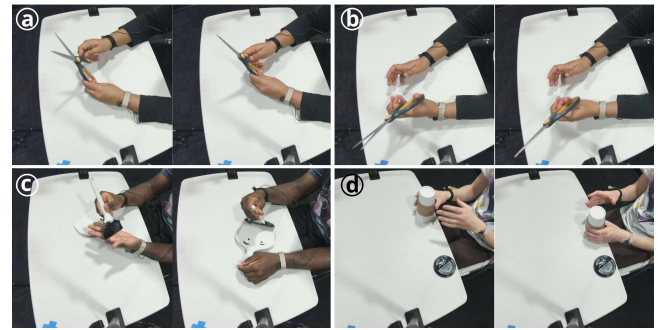


Figure 2: Examples of trial-and-error behavior from raw video footage of the study. (a) and (b): Open and close the scissors with different hands. (c) Brush on different surfaces such as the hand and table. (d) Knock the coffee cup on different parts of the table.

We found that participants who received audio augmentation during manual interactions reported a significantly greater sense of mindfulness than the *control group*. Through analyzing how participants described the objects they interacted with, we found that our device increased their attentiveness to the auditory properties of objects. *Device group* participants mentioned sound-related terms almost 10 times as the *control group* participants. This means that the auditory properties of objects usually go unnoticed during manual interactions, while our device renews attention to them. Behaviorally, we found that *device group* participants engaged with the objects for longer periods than the *control group*. They also exhibited more trial-and-error exploratory behaviors—repeated attempts with slight variations aimed at learning more about the objects, as shown in Figure 2. These patterns are manifestations of heightened curiosity [3, 14], prompted by our device, even when interacting with everyday, familiar objects.

Through this project, we demonstrated that subtle audio amplification of manual interactions can rekindle attention and curiosity

toward present-moment actions. This work is part of my broader effort to design technologies that support a sense of being “here.”

3 Sensory Scaffolding to Augment Social Interactions

Beyond manual interactions, “being here” also involves engaging with others through social interaction. These interactions are not limited to formal events but also about routine encounters embedded in everyday life. For instance, a quick greeting in a hallway or a shared moment on public transit can play a meaningful role in fostering a sense of social connection.

In these everyday encounters, we often form impressions based on physical adornments—such as clothing, hairstyle, or makeup—that signal aspects of personality, affiliation, or interest. However, these cues are relatively static and limited in their ability to convey dynamic, situational states, such as openness to conversation or emotional availability. In contrast, digital technologies provide affordances such as emojis, filters, and real-time status updates that allow users to actively shape how they are perceived. This contrast becomes especially pronounced in shared spaces, where people may be physically surrounded by others yet drawn toward digital media that offers more expressive social feedback.

This is where sensory scaffolding becomes valuable—as a form of temporary modulation of how one perceives another person, to augment face-to-face interactions.

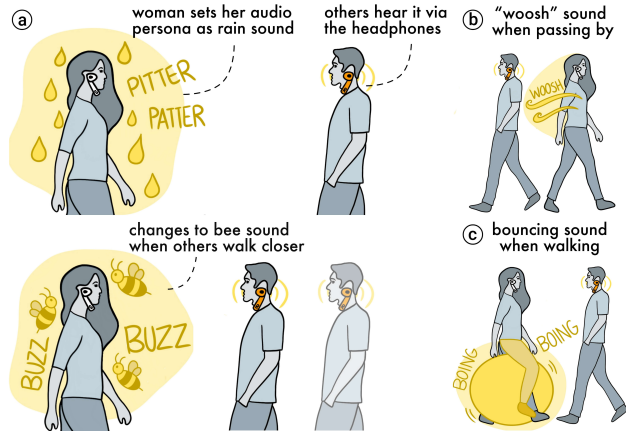


Figure 3: Demonstrating the concept of audio personas, body-anchored audio cues to enhance face-to-face social interaction. Here, the yellow illustrations indicate the content of the sound and are not rendered visually.

We introduce the concept of Audio Personas [24], which use sounds anchored to the body to augment social perception. Body-anchored audio cues create spatial alignment between the individual and the sound source. From a multisensory cue combination standpoint [23, 29], this spatial alignment increases the likelihood that the sound will be interpreted as relevant to that person. As a result, audio personas are designed to influence how one is perceived by adding an additional sensory layer to their presence.

Figure 3 visualizes a few examples of audio personas. For instance, one can choose buzzing bee sounds to indicate personal

boundaries, or a “woosh” sound to mimic passing by like a breeze for fun and playfulness. One could also customize their footsteps to sound like bouncing balls to express their love for gaming.

We conducted two user studies to explore the concept of audio personas (1) an in-lab study ($n = 64$) to examine whether audio personas are effective in influencing social perception, and (2) a design study ($n = 8$) to probe the use cases and design possibilities.

Audio personas are designed to be body-anchored. Therefore, in Study #1, as a between-subjects factor, we varied the spatial anchoring of sounds to assess the impact of this design choice, as shown in Figure 4. Participants interacted with two research assistants while hearing sounds either anchored to the assistants (i.e., audio personas) or to objects in the environment. As a within-subjects factor, we varied the valence of sounds to investigate how the emotional tone of audio personas influenced social perception.

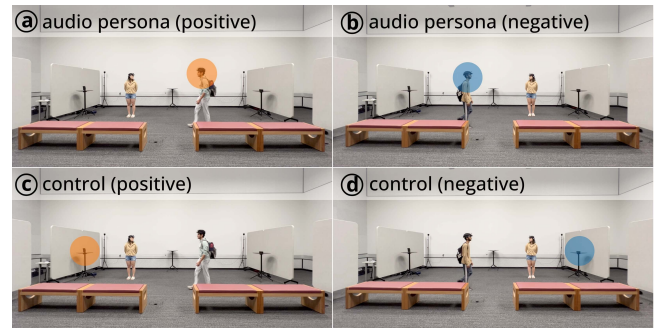


Figure 4: Condition design for in-lab study. (a) and (b) show the experience of the audio persona group, where the participants heard sounds anchored to the research assistant they interacted with. (c) and (d) show the experience of the control group, where participants heard audio anchored to objects in the space.

We found that audio personas are effective in shaping social impressions. Participants in the audio persona group were more likely to incorporate sound cues into their impressions of others compared to those in the control group. This was reflected in both their impression drawings and written descriptions. Moreover, the valence of the audio persona influenced the direction of the impression: individuals with positive audio personas were perceived as more socially attractive, likable, and less threatening than those with negative audio personas.

With Study #1 demonstrating the effectiveness of audio personas in shaping social perception, Study #2 explored use cases and design possibilities. We interviewed and co-designed with 8 participants experienced in audio design. We identified preferred use cases of audio personas in managing social impressions (e.g., personality & traits, identity & belonging) and signaling current states (e.g., emotions). Participants preferred using audio personas in public (e.g., street, transport) and semi-private settings (e.g., workspace, educational institutions) to form bonds with people they do not know well. We also elicited the creation of 48 audio personas from the designer participants and identified diverse design patterns in localization, triggers, dynamics, and content of audio personas.

Through Audio Personas, we demonstrated that sensory scaffolding can support “being here” in social interactions. By leveraging

spatialized sounds anchored to the body, these cues can shape users' social impressions and act as a support to kindle interactions in face-to-face contexts.

4 Sensory Scaffolding to Support Everyday Creativity (Ongoing)

In the previous sections, I highlighted how sensory scaffolding, the temporary modulation of perceptual experiences, can support a sense of “being here” in both physical and social interactions. My ongoing work extends the exploration to a higher-level construct that often emerges from these interactions: creativity.

While creativity is often associated with big, innovative ideas produced by exceptionally talented individuals, in everyday contexts, anyone can exercise creativity through small acts of problem-solving [2, 10]. Our ongoing project, in particular, focuses on the creative use of objects. This can be as simple as repurposing familiar objects for novel uses, such as rearranging furniture to better use a space or finding a substitute for a missing ingredient while cooking. In an era where computing systems are increasingly capable of automating problem-solving, these small, everyday moments of creativity become even more important for fostering genuine insight and deeper engagement with the ordinary.

To support everyday creativity, again, we tap into the perceptual process of how familiar objects are viewed and consider how this process can be supported through sensory scaffolding. In everyday interactions, we are frequently exposed to common use cases of familiar objects (e.g., chairs for sitting, cups for drinking). From a Bayesian perspective, this leads us to develop strong priors—expectations about how these objects should be used. While useful in many contexts, such expectations can lead to functional fixedness, particularly when users are challenged to repurpose objects for novel or alternative uses.

Building on these insights, we design sensory scaffolding systems that visually transform how users perceive everyday objects. For example, by exposing users with unconventional affordance of an object, the system aims to disrupt habitual perception. The goal is to scaffold the formation of new ways of seeing and reinterpreting familiar objects, thereby supporting their creative use.

5 Future Work

Through my Ph.D. research, I explore sensory scaffolding as a novel lens for supporting a sense of “being here” and fostering engagement with the physical, social, and creative processes of everyday life. Sensory scaffolding itself offers a rich space for further exploration, and below I outline several future directions.

5.1 Habituation and Long-term Impact

Sensory scaffolding systems temporarily modulate perceptual experiences to disrupt habituation and encourage renewed engagement with the ordinary. However, these systems themselves can become subject to repeated exposure, posing questions about their effectiveness over time. As part of the future work, I plan to study how users adapt to sensory scaffolding and design adaptive mechanisms.

For instance, using the previously described wrist-worn audio augmentation wearable as a platform, I plan to investigate how repeated exposure to enhanced auditory cues impacts attention

to those stimuli. By incorporating psychoacoustic methods, we will evaluate reaction times associated with responding to these cues, allowing us to assess how sustained attention develops over a duration of time. Response patterns can then be modeled to identify optimal moments for intervention.

Beyond controlled, in-lab methods for understanding how people adapt to sensory scaffolding, another important next step is to investigate the long-term effects of wearing such systems—extending beyond brief exposure to timescales of days, weeks, or even months. This is especially crucial for designing adaptive mechanisms that evolve over time, ensuring users continue to benefit from support without becoming over-reliant on the system. Ultimately, the goal of sensory scaffolding systems is to enable users to foster a stronger sense of presence in the ordinary on their own.

5.2 Safety and Ethical Considerations

Like other intervention systems, it is important to carefully consider potential risks introduced by sensory scaffolding during the design process. Sensory scaffolding is unique in that it directly interfaces with users' natural perceptual processes, making it essential to account for individual differences. For instance, people may have varying tolerance thresholds for audio amplification levels of their manual interactions, or they may interpret the same sounds differently based on cultural factors in the case of social signaling. Therefore, another line of future work on sensory scaffolding systems lies in personalizing the sensory cues delivered to individual users and finding optimal thresholds to achieve the intended effect.

Moreover, post-exposure effects must be carefully considered in the design of sensory scaffolding systems. How does altering the way people hear or see others influence their mental image of their own body and those around them? After users have experienced sensory scaffolding, how do they perceive the original, unmodified sensory cues from the real world? These are future work questions that could be grounded in existing literature on psychology and multisensory perception to ensure sensory scaffolding systems that offer meaningful support for everyday routines.

Last but not least, we acknowledge that wearable implementations of sensory scaffolding may constrain the benefits of the devices that can afford such devices, potentially exacerbating existing disparities in access to technology. Future work should explore alternative form factors for sensory scaffolding systems. For example, we envision sensory scaffolding integrated with the building around us—such as through ambient displays—to make these systems accessible to a broader range of users.

Acknowledgments

I would like to thank my Ph.D. advisor Sean Follmer and all the mentors and colleagues I have worked with in the aforementioned projects: Jeremy Bailenson, Libby Ye, Jingjin Li, Andrew Zhang, Jane L. E. and Ryan Tan. I would like to thank Han Guo for providing early feedback on this manuscript. I am supported by the Stanford Knight-Hennessy Scholarship.

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